Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Asia Pacific Square Mall II
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 8 – Decent Work & Economic Growth
Project / Solution Name	eViral - Your One-Stop Social Media Automation Platform.
Name (Leader)	Chin Hui Yan
Email & Phone (Leader)	chinhuiyan22@gmail.com 60162098926
University Affiliation(s)	Sunway University Asia Pacific University of Technology & Innovation (APU)
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Our team is tackling a problem for small and medium enterprises (SMEs) need social media to find customers, but most cannot afford the time or money to do it well. Malaysia has about 1.29 million MSMEs, making up 96.2% of all establishments (SME Corp / DOSM, 2025), and 80.3% of businesses already use social media (DOSM, 2024). Yet the people behind these businesses, often solo owners, home-based sellers and small food or service providers, spend an average of 6.3 hours a week on it, and 63% still say they do not have enough time. Owners rate their social media results just 4.2 out of 10, and 74% have no documented strategy, so posting stays inconsistent and content quality suffers.

This matters because the cost of getting social media wrong is lost revenue. With 76% of consumers saying they have purchased after discovering a product on social media (Forbes, 2025), an MSME that posts irregularly is missing sales it could have won. It also widens the gap between small businesses and large brands that employ full marketing teams. Beyond the economics, the burden falls on the owners themselves, who spend hours on content when they could improve their product, serving customers or spending time with family. This ties directly to SDG 8 (Decent Work and Economic Growth) and SDG 9 (Industry, Innovation and Infrastructure), with supporting links to SDG 1 and SDG 10 through more stable incomes and fairer access to digital tools.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

For our use case, artificial intelligence is useful for increasing relevance and outreach for a product. This is done through image recognition, which allows it to generate the appropriate hashtags and taglines for the postings. It also supports dual language caption generation, which helps target all users in Malaysia. While

yet to be implemented, we can use AI to web scrape and find current trends to follow to boost engagement and draw even more attention to the brand.

As for technical skills, the Replit workshop helped us by serving as an introduction to the platform, which allowed us to use it for ideation, making wireframes and deployment. The vibe-coding workshop helped us to prompt better to ensure that the AI actually follows the human design decisions instead of going off track.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

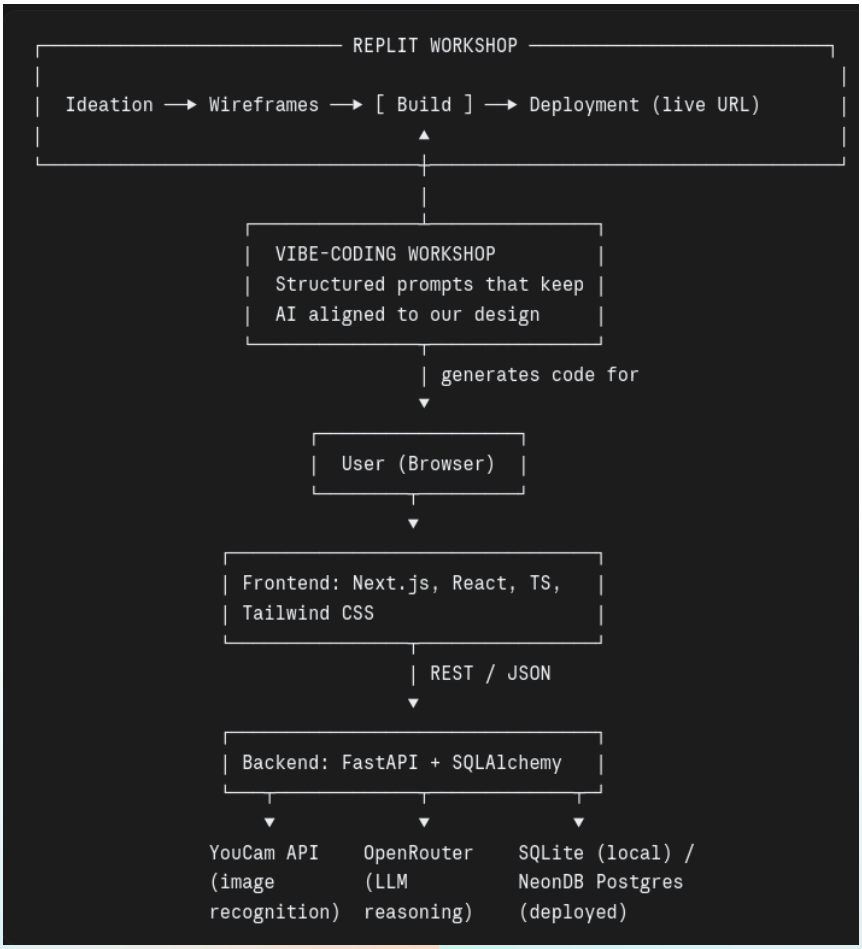
Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: Next.Js, TailwindCSS, ReactJS and Typescript

Backend: Python, FastAPI, SQLAlchemy, OpenRouter (LLM Reasoning) and YouCam API for image recognition.

Database: SQLite (Local), PostgreSQL (NeonDB for deployment)

Block Diagram Below:



3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Starting off, we focused on small business owners who manage social media themselves and may lack time or marketing expertise. We translated that problem into a workflow where an owner can provide a product name and photo, confirm the details the AI identifies, generate platform-specific captions, then review and schedule posts. This keeps the owner in control of product output while saving most of the work done.

We tested the idea through a working prototype. The prototype connects a Next.js frontend to a FastAPI backend, uses AI for product understanding and captions, and supports campaign review and scheduling.

We used a mock publisher to demonstrate the end-to-end flow without depending on live social-platform access.

For Phase 2, the prototype gives us a viable foundation, but deployment readiness still needs validation. We would need to test the workflow with target users, improve it based on their feedback, and complete production work such as social-platform authorization and publishing, deployment and security checks, and reliability testing.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 8.3, "Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises, including through access to financial services."
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	81% of small businesses fail to achieve consistent social media results, despite investing an average of 6.3 hours per week.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	% reduction in weekly time spent on social media management
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation*	60% reduction (from 6.3 hours to ~2.5 hours/week)

Metric Category	Your Response
e.g., "20% improvement in computational resource use."	

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next.js, React, Tailwind CSS, FastAPI, SQLAlchemy, Alembic	Open-source frameworks and code libraries	MIT
Other packages listed in frontend/package.json and backend/pyproject.toml	Open-source code libraries	Their individual licenses; see each package's license notice
NVIDIA Nemotron 3.5 Super VL; Google Gemma 4 31B; Qwen Qwen3.7 Flash, through OpenRouter	Pre-trained AI models	Model-specific terms. Gemma 4 31B is Apache 2.0; verify the terms for the model actually used, since eViral allows model selection and fallbacks. (Gemma model listing (https://openrouter.ai/google/gemma-4-31b-it), Nemotron model listing

		(https://openrouter.ai/nvidia/nemotron-3.5-super-vl))
YouCam Image API (youcam-image-v2), optional	Image-processing service/model	Proprietary service; subject to Perfect Corp's terms

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[CHIN HUI YAN]

[25/10/2026]